Egypt[31]

1 Introduction

On a per capita basis, Egypt's area of cultivated land at 0.05 ha per head is among the lowest in the world. Farm sizes are small with an estimated 70 percent of holdings less than 0.42 ha. Agriculture is almost entirely dependent on irrigation from the Nile River. The agricultural land-base of Egypt amounts to about 3.3 million hectares, consisting of 3.0 million hectares lying within the Nile basin and delta, and about 80 000 ha of oasis and rain-fed land. Of the total area in the Nile basin and delta, some 2.5 million hectares are "old" lands, and the remaining 0.72 million hectares are new reclaimed lands. In the old lands, an elaborate crop rotation system is followed. The main winter crops are wheat, berseem (Egyptian clover) and broad beans. Among the summer crops, maize, rice and cotton are dominant. Vegetable crops such as tomato, potato, cucumber, melons and others are cultivated in the three seasons.

Mixed farming is common, with a variety of crops being combined with a few heads of cattle, sheep or poultry. Most agricultural land is privately owned. Reclaimed new land, which was owned and operated by the government through public sector enterprises, has gradually been sold. Around 80 percent of this new land is currently operated by the private sector.

1.1 Importance of agriculture in the Egyptian economy

Table 1 presents some key indicators of the importance of agriculture in the Egyptian economy. The Egyptian economy has traditionally relied heavily on the agriculture sector as a source of growth and support for the non-agricultural sector. During the 1980s and 1990s, this dominance declined, but agriculture still accounts for a significant share of growth, exports and employment. The share of agricultural GDP fell from 29.3 percent in 1970 to 18.2 percent in 1980 and 16.6 percent in 2000. The share of agricultural employment declined from 35.8 percent in 1985 to 33.2 percent in 1990 and 28.1 percent in 2000. The relative decline in the role of agriculture partly reflects the strong growth in other sectors, particularly oil, services and construction. In addition, prior to the initiation of the sector policy reform in the 1980s, the sector was subjected to various distortions, which impacted negatively on its development. It also received a declining share of public sector investment during the last 25 years, which in turn was not always optimally allocated between the different subsectors.

Table 1. Key indicators of the importance of agriculture

	Unit	1990-94		1995-2000	
			Percent		Percent
GDP	US$ million	7 687	17.4	13 591	17.2
Population	million	24.8	42.3	25.0	38.6
Number of workers	thousand	4 525	33.6	5 069	28.19
Agricultural imports	US$ million	2 637	30.1	3 849	26.3
Agricultural exports	US$ million	426	13.4	528	10.8

Source: World Bank (2000), World Development Indicators; Ministry of Planning, Cairo, Egypt.

Egypt is a substantial net food importer. Rice is the only major food crop with an exportable surplus, but that surplus is relatively small and ranged between 3 and - 13 percent during the 1990s. The large increases in rice production were matched with equally large jumps in consumption with no significant increases in exports. All other commodities are imported with varying levels of self-sufficiency. Edible oil has the lowest self-sufficiency ratio as local production of that commodity satisfied only about 11 percent of the total consumption in the period 1998-1999.

1.2 Recent policy developments

Economic reform in the Egyptian agricultural sector was initiated in 1987 in the context of an SAP. The measures taken included liberalization of pricing and marketing of major crops, eliminating subsidies on agricultural inputs, liberalizing input markets, eliminating interest rate subsidies on agricultural loans and shifting from mandatory crop rotation to farmers' decision-based rotation. The second half of the 1990s witnessed liberalization of cotton marketing and trade and liberalization of the agricultural land rental market. Now, there are no government controls left in agriculture except the compulsory procurement of sugar cane at administered prices, the imposition of a maximum area for rice and geographical varietal distribution for cotton cultivation at the district level (Siam, 1999).

The SAP may have caused some negative impacts in terms of income distribution within the farming subsector as well as price and income instability, although it has led to a significant improvement in food self-sufficiency at the national level, which is an important element in food security. There has been an expansion of cereal acreage, wheat, maize and rice, at the expense of the cultivated area of cotton and green fodder crops. Cereal crops, particularly wheat, showed a significant improvement in yield per feddan. These improvements have been achieved through technological advances especially in the area of varieties that have been encouraged by incentives provided by SAP. Institutional reform in agriculture has lagged behind the financial and price reforms, which has impacted adversely on their performance. There has been some ambiguity with respect to the role of the state in a market economy as a new regime. So, in the transition period, 1987-1997, there has been a lack of institutional support in the areas of finance, R&D, agricultural extension, and marketing. The majority of Egyptian farmers have suffered from the adverse consequences of the institutional vacuum that occurred during the transitional period and still exists in one way or another.

In 1995, four items remained in the food subsidy system: bread, wheat flour, sugar and edible oil. Bread and wheat flour are available to all Egyptians without restriction, while sugar and edible oil are distributed monthly to consumers through ration cards. The number of ration cardholders has been reduced moderately; from 79 percent of the Egyptian population in 1994 to about 65 percent in 1999. As a result of its efforts to reduce the food subsidy, the government reduced the total budgeted cost of subsidies to LE4 billion, or around 1.5 percent of GDP, in 1999.

1.3 Participation in regional and multilateral integration efforts

Egypt is a party to a number of regional integration schemes. The European Mediterranean Agreement for Egypt was initialled in January 2001. In 1998, Egypt became a member of the COMESA, which opens up 19 new markets in Africa with more than 350 million consumers. Also, in January 1998, Egypt began implementing the Greater Arab Free Trade Agreement reached with Arab League members in connection with the Arab Common Market Treaty of the 1960s. Egypt has signed in addition a number of bilateral free trade agreements. Free trade zones have been created on a bilateral basis between Egypt and Tunisia, Turkey, Jordan, Lebanon, Morocco, Libya and Syria. Finally, in May 1998, Egypt and the United States agreed to begin talks on a Trade and Investment Framework Agreement, which is expected to be an intermediary step before strategy talks on a free trade agreement in the future.

Egypt has been, and continues to be, both a net agricultural importer and an NFIDC. A country's income category is a criterion for preferential treatment in the World Trade Organization negotiations. Countries are categorized in terms of income as low income,

lower-middle income, upper-middle income, and high income. Egypt belongs to the 52 countries in the lower-middle income group. Thus, while classified as a food-deficit country, Egypt is not in the low-income food-deficit group that is likely to capture most of the special and preferential treatment in the WTO negotiations.

2 Experience with implementing the WTO agreements

2.1 Trade policy prior to the Agreement

Since 1991, Egypt has taken aggressive steps to liberalize both domestic and foreign trade. Quantitative restrictions have been removed on imports, and tariff rates on most imported items have been reduced. With respect to exports, the process of exporting has been facilitated and simplified. From September 1996 to June 1997, non-tariff barriers on exports have been abolished. Administration procedures for exports are being streamlined.

Starting in May 1990, a number of commodities and commodity groups were taken off the list of prohibited imports, and with the continuous elimination of different items, the number of banned items was reduced by 1998 to 15, including poultry parts as well as certain textiles and apparel items (Khair El Din and El Shawarby, 2002).

As regards standards specification and quality control, the number of items that are subject to quality control was reduced by 1994 to 32 items. However, the list now consists of 183 items, as many products removed from the list of banned imports were placed on the quality control list. The restricted items include mainly foodstuffs. As illustrated in Table 2, more than 90 percent of food tariff lines are subject to quality control, compared with 50 percent for the overall economy.

Table 2. Number of food items and agricultural raw materials subject to quality control and bans

	Food	Agricultural raw materials	Overall economy
Total tariff lines	494	134	4 951
Lines permanently banned	16	6	231
Lines subject to quality control	446	45	2 509
Percentage of permanent bans	3.24	4.48	4.67
Percentage of lines under quality control	90.28	33.58	50.68

Source: Kheir El Din (2000).

In terms of non-tariff barriers on exports, the list of export prohibitions, which previously covered 20 commodities, had been considerably reduced by 1993, including only raw hides as well as waste paper and scrap metal. Quality control requirements were only maintained for foodstuffs.

2.2 Market access

As a result of the UR negotiations, Egypt presented base tariff rates for over 600 agricultural tariff lines and offered to bind all agricultural tariffs, compared with only 3 percent before negotiation. For most products, the bindings were in the range of 5-80 percent, with a commitment to their progressive reduction. It is estimated that the unweighted average of bound rates in 1998 was about 48 percent, down from 62 percent in the base period. It is expected to fall to about 28 percent by the end of the implementation period in 2005. The unweighted average of bound tariff rates for the all products (agricultural and non-agricultural) is expected to fall from 55 percent in the base period to 37 percent in 2005 (Table 3).

Table 3. Applied MFN tariff, bound and final tariff average for agriculture,

	Agriculture	Total

Percentage of lines (6 032)	5.2	100
Average applied (%)	18.5	26.8
Average including service fee and surcharge	21.8	30.2
Range (%)	1-40	0-3 000
Dispersion	73.5	475.6
Bound rates		
Base rate	62.2	54.9
Estimated rate, 1998	47.8	45.0
Final rate, 2005	27.5	37.1

Source: WTO (1999).

Applied tariffs have mostly been considerably below the bound rates. Applied tariffs exceed bound tariffs on less than 7 percent of agricultural items. The maximum tariff was gradually reduced during the pre-agreement period, from 100 percent in 1991 to 70 percent in 1994, and continued to be reduced during the post-agreement period down to 40 percent in 1998 with some exceptions such as alcoholic beverages and whole poultry. The resultant unweighted average of applied rates on all agricultural products (except alcoholic beverages) was estimated to be 18.5 percent (21.8 percent including service fee and surcharge), against the bound rate of 47.8 percent in 1998. Applied tariffs on major items such as wheat and maize were 1 percent in 2000. Similarly, tariffs on some major types of vegetable such as groundnut oil were very low (1 percent; Table 4).

Table 4. Bound and applied tariffs for selected agricultural products

Product	1995	2000		2005
	Bound (%)	Bound (%)	Applied (%)	Bound (%)
Poultry	80	70	70	60
Meat	5	10	5	10
Live fish	15	12.5	5	10
Fish, dried	50	45	30	40
Milk	40	35	30	30
Cut flowers	80	70	40	60
Potatoes seed	15	12.5	5	10
Lentils	5	5	1	5
Broad beans	5	5	1	5
Bananas	80	70	40	60
Citrus	80	70	40	60
Wheat	5	5	1	5
Barley	15	12.5	5	10
Maize (corn)	5	5	1	5
Rice	30	25	5[a]	20
Grain sorghum	15	12.5	5	10
Soya bean oil	20	17.5	15	20
Groundnut oil	30	25	1	20
Sugar beet (pure)	40	35	10[b]	30
Sugar cane (pure)	40	35	10[b]	30

[a] Reduced from 20 percent in 1999.
[b] Reduced from 24 percent in 2000.

Source: Ministry of Foreign Trade, Schedules of Concessions to WTO, Cairo.

The Government of Egypt (GOE) has recently introduced further measures to reduce restrictions and liberalize the trade regime. First, in 1998, GOE reduced the maximum tariffs on most products from 50 percent to 40 percent (except on poultry, alcoholic beverages, tobacco and cars) and consolidated rates of 35 percent and 45 percent to 30 percent. Second, it lifted the ban on most textiles in early 1998 in compliance with the WTO Agreement on Textiles and Clothing; the remaining import ban on apparel will be eliminated in 2003. Third, the tariff on rice imports was reduced from 20 percent to 5 percent. Fourth, the GOE has unified port inspections and testing in one entity to reduce the costs of importing and approved the acceptance of certificates of accredited laboratories abroad. These measures led to tariff equivalent reductions in the costs of importing. Presidential Decree 429 issued in 2000 has further modified tariffs on some items. Tariffs on raw sugar (cane and beet) for refining have been reduced from 24 percent to 5 percent. A tariff on pure sugar has been reduced from 24 percent to 10 percent. In addition, the government removed quantitative restrictions and many other non-tariff barriers to imports. Most of the tariff reductions that have been made since 1991 were in the context of the economic reform programme that was initiated in that year and continued throughout the 1990s. However, some specific product tariff reductions were made for reasons related to domestic price adjustments. In the case of sugar, for example, the GOE used tariff reduction to open up for more sugar imports in order to control the consumer price increase for this basic food product.

It can be concluded therefore that Egypt has had a mixed experience regarding market access during the post-agreement period. For many products, applied tariffs were consistently much lower than bound rates, suggesting that it was able to “live” comfortably with its relatively low bindings, although the consequences for trade flows (imports) are yet to be assessed fully. For several others, certain problems were encountered, e.g. on poultry and dairy products, delaying tariffication in one case and resulting in application of almost the full range of the bound duty in others.

As regards negotiations on further reductions in bound tariffs, Egypt may face some difficulties for many products because bound rates are low compared with those of many other countries. Since all countries face the same price fluctuations in world markets, those with lower bindings are more exposed. In a negotiating environment, higher benchmark rates would also have conferred some strategic advantage, e.g. in terms of greater leverage in market access negotiations. Lack of access to simpler safeguards such as those of the AoA is also a disadvantage, particularly when bound tariffs are low.

Attention can also be drawn to possible anomalies in Egypt’s tariff bindings. For instance, the rates are highest on poultry meat and citrus fruit and lowest on cereals, beef and sheep meat, whereas most countries have higher bound rates for “sensitive” import-competing products (poultry being an exception) and lower rates on export products (e.g. fruit). However, many other countries have higher bound rates for cereals as well as meat. Egypt may need to review its structure of bound tariffs to rationalize it from the standpoint of food security “sensitivity”.

In Egypt, Law 161 on “The Protection of the National Economy from the Effects of Injurious Practices in International Trade” was passed in 1998, together with its executive regulations. It introduced procedures to be followed regarding the application of safeguard measures as well as anti-dumping and countervailing duties. It was notified to the WTO in November 1998. According to this law, provisional safeguard measures may be imposed in the form of a tariff increase for a maximum period of 200 days. Definitive measures may be imposed in the form of quantitative measures or tariff increases or both. The measures are to be imposed to the extent necessary to prevent or remedy the injury caused. Definitive measures are applied for a period of up to four years and may be extended up to ten years (rather than the eight years provided for by Article 7 of the Agreement on Safeguards) inclusive of the period during which provisional measures are applied. In 2001, Egypt initiated a countervailing and safeguard case with regard to powdered milk, where a safeguard margin of 45 percent was imposed on imports of powdered milk.

2.3 Domestic support

The agricultural SAP implemented in Egypt has reduced government support for agriculture. Much of the support is currently provided in the form of general services. In the UR, Egypt did not provide details on domestic support measures, essentially on the ground that all support measures fell under one of the categories exempted from reduction commitments. In May 1999, for the first time, it notified support measures, for the period 1995-1998, in respect of Green Box and SDT outlays.

Total outlays on Green Box measures were put at about US$68 million in 1995, rising to US$76 million in 1996 but falling by half in 1997 on account of a sharp reduction of outlays on pest relief. Expenditure in 1998 plummeted to US$1.3 million, with the virtual ending of pest relief. As a result, 90 percent of the green box outlay in 1998 was related to relief from irrigation difficulties (Table 5). Compared with a value of agricultural output of about US$13 billion, these outlays were insignificant.

Table 5. Green Box outlays, 1995-1998 (US$ thousand)

Type of measure	1995	1996	1997	1998
General services	85	85	85	85
Pest relief	67 031	74 516	37 518	35 000
Relief from irrigation difficulties	1 177	1 177	1 177	1 177
Total	68 293	75 778	38 780	36 262

Source: Notification to WTO.

Table 6 shows outlays in the SDT category. They fell sharply in 1996, roughly stabilizing thereafter at about US$2.4 million. While fertilizer subsidies represented over 70 percent of total outlays in 1995, the total was evenly divided between fertilizer and seed subsidies thereafter. The AoA sets no limit on SDT outlays but requires them to be justified. In any event, the outlays are very small relative to the value of agricultural production, and Egypt would face little difficulty in complying with the provisions of the Agreement, even if these SDT expenditures were to be put into the non-product-specific AMS category.

Table 6. Outlays in the Special and Differential category, 1995-1998 (US$ thousand)

Type of measure	1995	1996	1997	1998
Input subsidy available to low-income producers				
Fertilizer	5 216	1 219	1 210	1 194
Seeds	1 917	1 227	1 227	1 218
Total	7 133	2 446	2 437	2 412

Source: Notification to WTO.

As regards trade-distorting support measures incorporated in the AMS, Egypt did not claim any in the base period; nor has it made a notification to the WTO so far. If Egypt should decide to provide such support in the future it could still do so as long as the outlays are limited to the *de minimis* level of 10 percent of the value of production of specific commodities (e.g. cotton and wheat) and the same percentage of the total value of agricultural production for non-product-specific AMS such as fertilizer and seed subsidies. It is thus highly unlikely that Egypt would be constrained by the current rules on domestic support measures.

The situation could change significantly if current policies on wheat, the staple food in the diet, were to change. It has been estimated that raising self-sufficiency for wheat from 48 percent to 60 percent would involve support exceeding the *de minimis* level (Lofgren and Kherallah, 1998). Limited land and water resources require that additional production of crops such as wheat must come from improvements in productivity. In the short run, this

may require increasing support for farm prices so as to encourage the adoption of new technology.

An issue of potentially great importance for Egypt, as well as for other countries of the region, is WTO rules governing capital investment in irrigation. Such expenditure is currently exempt from reduction commitments (Article 6.2). Adoption of advanced irrigation technology is vital to an increase in water supplies and improved efficiency of use. Ensuring that such investments continue to be exempt from reduction commitments is critical for the food security of these countries. To that end, Egypt needs to consider as a matter of priority the documentation and analysis of domestic support measures, both current and prospective. At the same time, it would be important to follow closely the debate on the status of investment subsidies in the new negotiations on agriculture.

2.4 Export subsidies

Since it did not declare any export subsidies in its WTO Schedules, Egypt is not eligible to provide such subsidies in the future. In practice, this limitation is unlikely to be of any consequence, as export subsidies are neither desirable nor affordable for most developing countries. Moreover, the predominant feeling in WTO seems to be either to prohibit the subsidies completely or to restrict them severely. Under current rules, Egypt would still be able to provide subsidies to lower internal transport and marketing costs and external freight costs. A Fund for Equalizing Import and Export Prices was established and has been supported from the government by LE400 million aiming at developing production practices for improving export capability. This fund is distributed approximately 40:60 between the agricultural and textile sectors.

Egypt also applies other incentive measures aimed at export promotion. For example, customs duty can be reduced on a selective basis to encourage greater local content and export-oriented activities. Similarly, the Export Development Bank of Egypt provides short- and medium-term loans to finance capital assets of export-oriented industries and credit to finance inputs for these industries. Processed agricultural products are among the beneficiaries of such credits.

Export prohibitions that formerly prevailed for some agricultural products have mostly been eliminated. For example, the ban on tanned leather exports was lifted in 1994, and that on raw hides in 1998. Previously, there were export quotas on wool, wool waste, cotton waste and tanned leather, but they were removed in 1993. A decision was taken recently to restrict the export of “baby” potatoes unless grown in specified fields. This was said to be related to EU technical regulations on imports of potatoes.

2.5 State trading enterprises

In Egypt, STEs play a major role in wheat imports and cotton exports. Egypt is one of the largest importers of wheat in the world, and the General Authority of Supply Commodities (GASC) imports about three-quarters of these wheat. GASC is an agency located in the Ministry of Supply and Home Trade, and is designated to carry out the food subsidy programmes in Egypt. It deals with wheat, sugar and edible oils, which are the commodities included in the subsidy programme. GASC bears the responsibility of purchasing domestic wheat as well as wheat imports both used for producing 82 percent flour. This is used in baking subsidized “baladi” bread. In the case of cotton, the public sector companies export three-quarters of the total Egyptian cotton exports. Other cotton products, such as yarn, fabrics and garments, are also largely exported or imported by public sector companies (STEs). Egypt does not have any import cartels.

2.6 SPS and TBT Agreements

Egypt has gained considerable post-agreement experience in this area. In view of the growing importance of these Agreements for Egypt, a summary of this experience is provided below.

First, there have been two SPS-related experiences regarding the importation of poultry products. One concerned a decree banning the importation of poultry parts because it was

difficult to ascertain whether or not the imported parts came from poultry slaughtered in accordance with Islamic traditions, i.e. Halal. The second related to the specification of a maximum moisture content of frozen poultry of 5 percent, which was considered by some WTO members to be well below the average moisture content permitted in many other countries.

Second, some WTO members sought clarification in December 1997 of the basis for, and application of, the prohibition of beef imports with a fat content greater than 7 percent and how this restriction could be justified under the national treatment provisions of GATT Article III.

Third, the Ministry of Agriculture restricts all imports of cotton to specified port areas as a precaution against the introduction of potentially harmful pests and diseases.

Fourth, in June 1999, the Ministry of Trade and Supply issued a new Decree banning the importation in that year of some food products from the EU that might be contaminated with dioxin. The ban affected mainly meat, egg and dairy products.

Fifth, concerning fruit and vegetable exports to the EU market, there have been no particular issues. It is generally felt that most of the SPS measures applied by EU on these products are justified. In fact, one study found that a group of EU traders surveyed considered that measures applied by many other countries were more restrictive than those of EU. One exception to this generalization is related to a change in the rule (towards a stricter import regime) following the establishment of the EU single market in 1993. With the single market, EU-wide standards were raised in 1998 to protect the Union's southern members, which essentially meant that exporters faced higher standards overall. While this may be WTO-compatible, a grey area for consideration was: Why would Italy allow the import of citrus from an area infected by white fly only in periods of domestic market shortage if the danger of infection were that significant? This may be an issue for Egypt to pursue. Australia and China also prevented Egypt's citrus exports for the same reason of white fly infection.

Sixth, a decision was taken recently to restrict the export of "baby" potatoes unless grown in specified fields. As noted above, this was said to be related to EU technical regulations on imports of potatoes.

Seventh, two Ministerial Decrees have been issued, aiming at ensuring good production techniques and biosafety to ensure access to importing countries' markets. The Ministry of Agriculture and Land Reclamation and the Ministry of Economy and Foreign Trade (now the Ministry of Foreign Trade) have already issued the organizing rules for two major export commodities, potatoes and groundnuts. Similar Ministerial decrees are being prepared for vegetables (beans, onion and garlic); fruits (grapes, citrus, strawberry and cantaloupe) and cut flowers. These decrees specify the locations that are permitted to grow exportable crops (to ensure the products are free of pests) and determine types and sources of seeds as well as methods of seed treatment, pest control methods for crops and soil. These specifications are prepared in consultation with importing countries.

Egypt has notified WTO about the technical assistance requirements in the area of SPS. Egypt is asking for technical assistance and advice in the areas of food safety, animal health and upgrading human skills in the areas of rights, obligations, and practical operation of the SPS Agreement.

Compliance with sanitary and phytosanitary requirements needs certain investments. For Egypt, an example would be the cost to achieve brown rot-free status to export potatoes. In the case of horticultural exports, the application of Good Agricultural Practices requires considerable investments in training and improved extension services. Furthermore, particularly in the case of processed export goods, improvements and investments in hygiene standards might be required throughout the sector, i.e. not only in processing plants but also on farms themselves and in transportation. The necessary resources to develop a system to monitor and enforce/comply with sanitary and phytosanitary requirements will be substantial, both financially and technically.

2.7 TRIPS Agreement (agricultural aspects)

The TRIPS Agreement leaves member countries free to devise protection for plant varieties by means of patents, by an “effective” *sui generis* regime or by a combination of both modalities (Article 27.3 [b]).

An IPR law has been issued by the People’s Assembly Parliament of Egypt. Chapter 4 in the law deals with plant variety protection. The essentials of the manner in which plant varieties are protected derive extensively from the UPOV Convention (1991). All farmers are granted the right to use saved seeds for replanting on their own lands. This allows farmers to maintain their traditional practice of saving seeds without paying royalties, even when they are using seeds of new varieties developed and protected by local researchers. The law gives the authority to the Minister to decree which crops can benefit from farmers’ privilege.

However, for the Egyptian law to conform to the UPOV, it does not allow farmers to multiply and sell seeds of protected varieties without paying royalties, i.e. it does not allow farmers to set up a small seed-pirating business. Note that current Egyptian law prohibits the sale of seeds without a commercial licence (Law 53/1966, art. 56; Kent, 2000). The Council of UPOV has examined the draft Egyptian law and the corresponding Decree, and found that it would bring the Egyptian legislation on the protection of plant varieties fully into conformity with the 1991 Act of the UPOV Convention.

Egypt has an urgent need for training in virtually all IPR fields and for human resource development. Accordingly, Egypt is asking for technical assistance and advice in the areas of amending laws and regulations, upgrading judicial and administrative frameworks, building human resource capacity, collecting and disseminating technical information about IPRs, and strengthening negotiating capacities for technology acquisition.

In terms of the potential impact of the implementation of intellectual property rights on Egyptian agriculture, mixed results can be expected. IPR generally is essential for making substantial technological advances in agriculture, and these are becoming more important in particular, given the limited agricultural resource base. As a good example, Egypt has used high-yield varieties of cereal crops to make a substantial increase in food production during the last 15 years. However, implementation of IPR in the context of Egyptian agriculture could have an adverse impact on farmers’ well-being. With higher costs of seeds of high yielding varieties (HYV) under plant varieties protection, small farmers (the majority of farmers in Egypt) will have less access. Differential access between small and large farmers is likely, leading to increased income inequalities and endangering food security at the household level.

Also, improved varieties could be monopolized by large companies and multinationals, which - intending to make huge profits - would raise prices of new technologies to unaffordable levels with respect to small farmers who are usually in a weak financial situation.

In Egypt, although the intellectual property rights (IPR) law has already been issued, institutions that are essential for the implementation of this law in agriculture have not yet been established. Institutional capacity building in the IPR areas will take a long time. There is a need for large investments as well as technical assistance. For these reasons, the issue of IPR should be an urgent priority, especially for a developing country like Egypt, to protect food security and to ensure uninterrupted supply of new technologies to resource-poor farmers (Lele *et al*., 1999).

2.8 Impact of agreements on export market access opportunities

Egypt receives OECD trade preferences under the GSP that allow its exports access to these markets at zero duties or at tariffs below MFN rates. The UR will reduce OECD countries’ MFN tariffs by 40 percent on average. A reduction in MFN tariffs will reduce, or even eliminate, these preference margins.

About 15 percent of EU imports from Egypt gain GSP treatment, while the corresponding shares to Japan and the United States are lower than 4 percent. Japan's largest imports from Egypt are raw cotton and crude petroleum, which are imported under zero MFN tariffs. Table 7 shows that GSP is either not extended to Egypt's most important export products or could not be extended, since these items already face zero MFN duties. Therefore, erosion of Egypt's preferences under GSP may not be very significant. The net gains that Egypt might achieve depend on the ability to offset losses owing to erosion of preference margins and on its capacity to improve the competitiveness of its products. A study undertaken jointly by the World Bank and the United Nations Conference on Trade and Development (UNCTAD) suggests that UR liberalization would produce only minor gains for Egypt, as gains from MFN tariff reduction will be almost offset by losses owing to preference erosion (Kheir El Din, 2000).

Table 7. Share of Egyptian exports receiving OECD preference

Import market	Percentage share of Egypt's exports under different tariff regimes			
	Zero MFN tariffs	Zero GSP rate	Non-zero GSP rates	Non-zero MFN rates
EU	58.3	14.4	0.2	27.1
Japan	40.8	3.7	0.1	55.4
United States	12.0	3.9	0.0	84.1

Source: Kheir El Din (2000).

As for NTBs imposed especially on exports of agricultural products, the UR made important progress in liberalizing them. It has been estimated that about 17.6 percent of all Egyptian exports to the United States face NTBs, while the ratio in the EU is 13.2 percent. Foodstuffs exported to the EU face several different types of restraining measures. Post-agreement average ratios will fall from their aforementioned current levels to 1 percent in the EU and 2 percent in the OECD countries owing to the removal of NTBs applied to agricultural products, textile and clothing (Kheir El Din, 2000).

Bringing textiles and clothing trade under WTO rules is generally a positive development for developing countries as a whole as a result of the elimination of NTBs applied under the Multi-Fibre Agreement (MFA). However, there may be negative implications for individual exporters of MFA products. For Egypt, as a small producer, assigned quotas may be considered as a means of guaranteed access. Phasing out of the MFA quotas on textile and clothing trade will mean that Egyptian exporters face increased competition from other large and possibly more efficient suppliers whose trade has been constrained by MFA restrictions (Khair El Din, 2000). The expected trade diversion away from Egypt towards other competitors will be much larger than any potential increase Egypt could achieve in the markets of textile and clothing products, unless the Egyptian textile industry achieves a greater efficiency in production. The prices of these products will decrease also as a result of the elimination of distortions created by the MFA. For cotton, however, the elimination of the implicit tax on cotton producers under the MFA will result in an increase in cotton prices in world markets relative to prices of man-made fibres. This will divert the textile industry away from cotton towards a greater use of synthetic fibres. Therefore, in the long run, Egypt may suffer a net loss from the replacement of the MFA by WTO rules.

3 Evaluation of trends in food and agricultural trade

3.1 Macroeconomic background

In 1991, Egypt started to implement a comprehensive economic reform programme that successfully achieved macroeconomic stability. Central to this success were three elements: a major fiscal adjustment, exchange market reform and the adoption of an exchange rate anchor, and a supportive monetary policy. In the period 1981-1986, just before the reform policy in agriculture, the exchange rate was overvalued by 95 percent. In the period 1986-1989, the exchange rate was overvalued by over 200 percent, falling to 47 percent in 1990 and reaching parity in 1991.

In October 1991, segmented markets for foreign exchange were unified at a value determined by market forces (Kheir El Din and El Shawarby, 2002). The nominal exchange rate (NER) was devalued by 23 percent from US$1 to LE2.708 at the end of June 1990, to LE3.342 by February 1991. Since then, the NER of the Egyptian pound *vis-à-vis* the US dollar, used as a nominal anchor, remained roughly constant, although this resulted in a steady real appreciation, which amounted to about 40 percent by 1998. The currency was devalued again to 3.98LE/US$ in 2001 and by a further 30 percent to 4.6LE/US$ in the first quarter of 2002.

Because of this devaluation, imports are now more expensive in local currency, and Egyptian exports should be more competitive in world markets. However, devaluation has adverse effects on the food security situation, as imported food commodities are now more expensive. Whether a nominal or a real depreciation can help net exports in the Egyptian case is controversial. In this regard, a study by El Shawarby (1999) about the likely impact and effectiveness of devaluation on Egyptian exports emphasizes the evidence on the weak role of exchange rate variations on Egyptian export performance. The study concludes that export promotion in Egypt is not entirely an exchange rate problem, but rather a behavioural and institutional problem.

3.2 Overall balance on agriculture and food trade

In the period 1985-1989, the total agricultural trade averaged US$3.81 billion annually, representing about one-third (34.1 percent) of Egypt's trade, but decreased in both absolute and relative terms to average US$3.06 billion in the period 1990-1994, representing 25.7 percent of Egypt's trade. In the latest period 1995-2001, although the total agricultural trade increased to US$4.38 billion, its share decreased further to 22.4 percent of total trade.

As Table 8 shows, agricultural exports decreased from US$610 million on average in the period 1985-1989 to US$457 million in the period 1993-1994 and made a slight increase in the period 1995-2001 to an average of US$528 million. The share of agricultural exports in total exports decreased from 22.7 percent in the first period to 13.4 percent in the second period and to 10.8 percent in the third period.

Agricultural imports decreased between the first two periods from US$3.20 billion to US$2.64 billion but increased to US$3.85 billion on average in the third period with a share of 37.8 percent, 30.1 percent and 26.3 percent of total imports in the three periods, respectively. The terms of trade index, which reflects the ratio of export unit value to import unit value, shows a decreasing trend during the period 1995-2000 compared with the period 1990-1995.

Egypt continues to record a large agricultural trade deficit, which grew from US$2.4 billion in 1985-1994 to over US$3.32 billion in 1995-2001.

Table 8. Agriculture and food trade (annual averages)

Period	Imports		Exports		Net exports	
	Agriculture	Food	Agriculture	Food	Agriculture	Food
	US$ million *per annum*					
1985-94 (A)	2 919	2 285	518	238	-2 402	-2 047
1995-2000 (B)	3 537	2 671	526	315	-3 010	-2 357
1995-2000 (C)	2 154	1 612	319	242	-1 834	-1 370
	Growth rate (%) *per annum*					
Period A	-4.0	-4.0	-6.0	5.0	-	-
Period B	0.2	0.5	0.7	-3.0	-	-

Note: Period C shows what average annual imports and exports would have been if the trend in the 1985-1994 period had continued.

Source: Calculated from FAOSTAT.

Food imports accounted for roughly 78 percent of all Egyptian agricultural imports on average in the period 1985-1989. In the post-agreement period (1995-2000), the share of food imports continued to decrease, reaching 70 percent, even though the value of food imports increased to US$2.67 billion on average in this period. By contrast, the share of food products in total agricultural exports has increased, from about 46 percent in the period 1985-1994 to about 60 percent in the period 1995-2000. Net food imports amounted to US$2.34 billion in 1985-1989 and increased slightly in 1995-2000 to a value of US$2.361 billion.

3.3 Trends in agricultural exports

Egypt traditionally has had a concentrated export profile, but it is becoming more diversified over time. In the period 1985-1989 agricultural exports were dominated by cotton, which constituted about 60 percent of the total agricultural exports. Orange exports amounted to a further 10.5 percent of the total. In the pre-agreement period 1990-1994, cotton's share dropped to only 27 percent. Exports of rice, potatoes and oranges together constituted 25.7 percent of the total. In the post-agreement 1995-2001 period, exports of cotton have stabilized at 30.0 percent, while the share of rice, potatoes and oranges grew further to 37.1 percent. There was also a corresponding drop in the contribution of "other" exports from 43.9 to 30.2 percent (Table 9).

Table 9. Agricultural exports by commodity

	Period averages						Annual percentage change	
	1985-89		1990-94		1995-2000			
	(A)	Percentage of total	(B)	Percentage of total	(C)	Percentage of total	B over A	C over B
Cotton								
Unit value (US$/tonne)	364	59.6	115	27.0	158	30.0	-20.6	6.6
Value (US$ million)	112		40		67		-18.6	10.9
Quantity (thousand tonnes)	3 250		2 873		2 358		-2.4	-3.9
Rice								
Unit value (US$/tonne)	17	2.8	45	10.6	95	18.1	21.5	16.1
Value (US$ million)	52		155		296		24.4	13.8
Quantity (thousand tonnes)	327		290		321		-2.4	2.1
Potato								
Unit value (US$/tonne)	26	4.3	34	8.0	56	10.6	5.5	10.5
Value (US$ million)	136		174		284		5.1	10.3
Quantity (thousand tonnes)	191		195		197		0.4	0.2
Onion								
Unit value	10	1.6	15	3.5	14	2.7	8.4	-1.4
(US$/tonne)								

Value (US$ million)	35		89		121		20.5	6.3
Quantity (thousand tonnes)	286		169		116		-10.0	-7.2
Oranges								
Unit value	64	10.5	30	7.0	44	8.4	-14.1	8.0
(US$/tonne)								
Value (US$ million)	119		88		160		-5.9	12.7
Quantity (thousand tonnes)	538		341		275		-8.7	-4.2
Other products								
Value (US$ million)	129	21.1	187	43.9	159	30.2	7.7	-3.2
Agricultural exports (US$ million)	610	100	426	100	526	100	-6.9	4.3

Source: Calculated from Egypt CAPMAS.

Egypt's export profile is concentrated not only in terms of commodities but also in terms of markets. The EU is the largest market for Egyptian exports with a share of 44 percent of the total in the post-agreement period, while the United States has the second largest share, 13 percent. Exports to Arab countries represent 10 percent of total Egyptian exports. The EU was the most important destination for Egypt's exports of cotton and potatoes, with 24.5 and 80.4 percent, respectively, in the pre-agreement period 1993-1994 and 36.2 and 81.4 percent in the post-agreement period 1995-2001, respectively. It seems also that the EU opened up its markets for Egyptian exports of cotton and potatoes in the postagreement period in both absolute and relative terms; the volumes are greater, and the shares are larger. However, the EU is of minor importance as a market for Egypt's rice, oranges and onions. Arab countries in general and the Gulf countries in particular are important markets for Egypt's exports of rice and oranges and, to a lesser extent, potatoes and onions.

Egyptian exporters still face serious constraints on increasing sales abroad. Domestically, the constraints include: low-quality domestic input, cumbersome duty - drawback and temporary admission regimes, excessive paperwork, fees and delays for customs and various inspections during export and import; workers that are poorly prepared for the jobs available; insufficient incentives to export; and lack of access to information on foreign markets and product standards.

3.4 Trends in agricultural imports

Egypt's agricultural imports have expanded significantly between the pre- and post-agreement periods; from US$2.6 billion on average in the period 1990-1994 to US$3.5 billion on average in the period 1995-2000 with an annual rate of increase of 5.9 percent, as shown in Table 10. Wheat, maize, edible oil, sugar, powdered milk and red meat are Egypt's major import commodities. Valued at US$767 million, wheat imports (including grains and flour) constituted 29.0 percent of the total agricultural imports in the period 1990-1994 but decreased to 22.9 percent with a value of US$803 million on average in the period 1995-2000. In terms of maize and edible oil, their imports increased in both absolute and relative terms; the value of maize imports increased from US$227 million with 8.4 percent the first period to US$462 million representing 12 percent in the period 1995-2001. The share of edible oil imports increased from 8 percent to 9.9 percent.

Table 10. Agricultural imports by commodity

	Period averages						Annual percentage change	
	1985-89		1990-94		1995-2000			
	(A)	Percentage of total	(B)	Percentage of total	(C)	Percentage of total	B over A	C over B
Wheat								
Value (US$ million)	1 113	34.8	767	29.0	803	22.9	-7.2	0.9
Quantity (thousand tonnes)	6 904		5 695		4 809		-3.8	-3.3
Unit value (US$/tonne)	161		135		167		-3.5	4.3
Maize								
Value (US$ million)	214	6.7	240	9.1	449	12.8	2.3	13.3
Quantity (thousand tonnes)	1 774		1 800		3 266		0.3	12.7
Unit value (US$/tonne)	121		133		137		1.9	0.6
Sugar								
Value (US$ million)	174	5.4	179	6.8	210	6.0	0.6	3.2
Quantity (thousand tonnes)	670		543		707		-4.9	5.4
Unit value (US$/tonne)	260		330		297		3.7	-2.1
Edible oils								
Value (US$ million)	199	6.2	231	8.7	416	11.9	3.0	12.5
Quantity (thousand tonnes)	346		538		705		9.2	5.6
Unit value (US$/tonne)	575		429		590		-5.7	6.6
Red meat								
Value (US$ million)	271	8.5	138	5.2	196	5.6	-12.6	7.3
Quantity (thousand tonnes)	202		133		156		-8.0	3.2
Unit value (US$/tonne)	1 342		1 038		1 226		-5.0	3.4
Milk equivalent								
Value (US$ million)	210	6.6	153	5.8	166	4.7	-6.1	1.6
Quantity (thousand tonnes)	817		601		594		-6.0	-0.2
Unit value (US$/tonne)	257		255		279		-0.2	1.8
Other products value (US$ million)	1 017	31.8	933	35.3	1 270	36.2	-1.7	
Agricultural imports (US$ million)	3 198	100	2 641	100	3 510		-3.8	

Source: Calculated from Egypt CAPMAS.

Egypt depends on several sources for supplying agricultural and food imports. However, there has been some concentration in terms of the origins of each food commodity. The

origin of Egyptian imports of cereals in particular is almost exclusively the United States, which supplied Egypt with 65 percent of wheat and 77 percent of maize (yellow corn for poultry feed) in the pre-agreement period (1993-1994), and 67.7 percent of wheat and 81.7 percent of maize in the postagreement period (1995-2001). Australia is the second supplier but with a smaller share in the second period. Malaysia is the main origin of Egypt's imports of edible oil: 49 percent in the first period and 44.8 percent (albeit with a greater volume) in the second period. As for sugar, Cuba has been overtaken by Brazil as the main supplier of Egypt's sugar imports. In the second period, Brazil supplied Egypt with more than half of sugar imports, compared with only 24.4 percent in the first period.

4 Food security impacts

4.1 Trends in food security indicators

This section addresses the impact of agricultural trade liberalization on food security in Egypt by analysing the trends in selected food security indicators. Trends in these indicators for the pre- and post-agreement periods (1985-1994 and 1995-1999) are summarized in Table 11.

The number of calories per capita available in Egypt according to FAO data seems very high. The daily calorie supply per capita, as indicated in Table 11, amounted to 3 211 kcal on average for the period 1990-1994, which is 30 percent higher than average energy requirements and higher than levels observed in most developing countries. The per capita energy intake for the post-agreement period (1995-1999) is even higher and is estimated at 3 305 kcal or about a 3 percent increase. About 93 percent of this energy is supplied from vegetable products, and the remaining 7 percent originates from animal products. The daily protein supply per capita is 85 g in 1990-1994, with a slight improvement in the period 1995-2000 of 90 g. However, the animal protein component is as low as 15 percent of daily protein per capita, which indicates the existence of a quality problem, rather than a quantity problem, in the Egyptian diet.

Total food availability, the sum of food production and food imports, increased from 40.9 million tonnes on average in the period 1985-1989 to 46.2 million tonnes on average in the period 1990-1994 and to 58.7 million tonnes on average in the period 1995-2000. The share of domestic food production in total food availability increased from 72.6 percent in the first period to 77.6 percent in the second period and then to 78.0 percent in the third period.

Taking the period 1989-1991 as a base, the food production index increased from 115 in 1994 to 154 in 2000. The food production per capita index increased from 106 in 1994 to 127 in 2000.

Table 11. Evolution of selected food security indicators

Indicator	Average 1985-89	Average 1990-94	Average 1995-99
Diet			
Total energy intake (kcal)	3 093	3 211	3 305
Percent sources			
Vegetable products (%)	92.4	93.5	93.1
Animal products (%)	7.6	6.5	6.9
Protein intake (g)	80.78	85.4	90.7
Percent sources			
Vegetable products (%)	83.7	84.7	82.6
Animal products (%)	16.2	15.3	17.4
Food supply			
Food production (thousand tonnes)	2 961	35 810	45 798

Food imports (thousand tonnes)	11 223	10 344	12 888
Total supply (thousand tonnes)	40 894	46 154	58 686
Food production to total (%)	72.6	77.6	78.0
Food production index, 1989-91 = 100	n.a.	109.2	142.5
Food production per capita index, 1989-91 = 100	n.a.	104.6	123.3
Price indexes (1990 = 100)			
Consumer price index	n.a.	139	221.6
Food price index	n.a.	125	178.8
Non-food price index	n.a.	161.6	258.5
Ratio food to non-food price indexes	n.a.	77.4	69.2
Food import capacity ratio	n.a.		
FICR1[a]	n.a.	0.17	0.17
FICR2[b]	n.a.	0.55	0.52
Agricultural tradability index	n.a.	0.4	0.3

n.a.: not available.

[a] FICR1: Food import capacity ratio defined as the actual value of food imports to total export revenues of goods and services.

[b] FICR2: Food import capacity ratio defined as the actual value of food imports to export revenues of merchandise only.

Source: Calculated from FAO. FAOSTAT, and World Bank, World Development Indicators and African Development Indicators.

Regarding growth rates of yield per feddan for the major crops, the first period (1985-1994) coincided with the implementation of the first phase of the structural adjustment programme in agriculture (1987-1997) as discussed before, whereas the post-agreement period (1995-2001) coincided with the second phase of SAP as well as the implementation of UR. Growth in productivity has varied among crops. Cereal crops have achieved high rates of growth in productivity during the period 1985-1994. Wheat showed the best performance in this group of crops, with an annual rate of growth at 3.34 percent, followed by rice (4.06 percent) and maize (3.58 percent). In contrast, cotton productivity grew slowly, achieving only a 0.69 percent growth rate. Growth rates with respect to cereals productivity in general and wheat and rice in particular have slowed down in the post-agreement period, 1995-2001.

Nominal food prices in Egypt have increased regularly throughout the 1980s and 1990s. The food price index (with 1990 = 100) reached 190 in 1999, which means that domestic food prices almost doubled during the 1990s. This compares with an increase of more than five times during the 1980s where the food price index was only 18 in 1980. Despite the increase in domestic food prices in nominal terms, the ratio of food to non-food prices indexes decreased from 77.4 percent on average in the period 1990-1994 to 69.2 percent on average in the period 1995-1999, indicating a substantial reduction in the real price of food. While this benefits the urban and non-farm rural population in terms of nutritional food security, farm population food security is harmed as a result of the deterioration of their real income.

The food import capacity ratio - defined as the value of food imports to total export revenues (merchandise only) - indicates the demand for foreign exchange to finance food imports. As shown in Table 11, this ratio for Egypt is relatively high. Relatively little change occurred in this indicator between the pre- and postagreement periods. The high dependence on food imports makes Egypt vulnerable, particularly in years of domestic harvest shortfalls or higher world prices, when this ratio can increase substantially.

The ratio of trade to GDP is used as an indicator for openness and vulnerability of the economy. The same indicator can also be applied at the agricultural sector level. An agricultural tradability index can be defined as the ratio of the sum of agricultural exports plus imports relative to agricultural GDP. This indicator reflects the sensitivity of a country's agricultural sector to price developments in world markets. The agricultural tradability index for Egypt decreased from 0.4 on average for the period 1990-1994 to 0.3 on average for the period 1995-2000.

Two important issues related to the agricultural tradability index are the extent of concentration in trading partners and the degree of concentration in the commodities exported. With very concentrated trade profiles, Egypt is more vulnerable to unilateral trade measures and possibly more restricted in its negotiating powers.

4.2 Trends in poverty and undernutrition

Poverty in Egypt's major social problem. About one-quarter of the population are poor in absolute terms, with women and children among the most vulnerable groups. Despite Egypt's sophisticated social protection system, poverty is widespread, and social risks are still a major contributory factor to poverty in Egypt. Using consumption expenditure-based poverty lines, the 1995/1996 estimated poverty measure indicated an overall headcount index of 22.9 percent, which was slightly higher in rural areas. Measuring poverty over the period 1981-1996, one study (Osman, n.d.) indicated a rise in poverty in urban Egypt, while the headcount measure is shown to have declined in rural Egypt.

In the rural areas, the poor are mainly farmers with little or no land as well as agricultural labourers. There is a smaller group of ultra-poor, 13 percent in rural areas, including widows, sick and disabled older people, heavily dependent on direct income transfers. The largest concentration of poor and ultra-poor individuals is found in upper Egypt in both urban and rural areas. In 1994, an International Food Policy Research Institute (IFPRI) study revealed that poverty had increased slightly during the period 1981/1982-1990/1991 in both urban and rural areas. A more recent IFPRI (1999) study reveals that, in 1997, 25 percent of the subsample households were below the poverty line. The incidence of poverty for this sample had increased to 33 percent in 1999. Moreover, almost half of all poverty is persistently chronic (that is, the average consumption over time is below the poverty line). The study also revealed that per capita consumption had decreased for the sample households from an average of LE231 per capita per month in 1997 to LE213 per capita per month in 1999 (Haddad and Ahmed, 1999).

Malnutrition remains a serious health problem in Egypt where under- as well as overnutrition are found. Chronic undernutrition (stunting) and micronutrient deficiencies are most prevalent in rural areas. About 17 percent of the under-five population is underweight, 22 percent stunted and 9.7 percent wasted. Irondeficiency anaemia is a common problem, goitre is endemic, and rickets and other nutritional deficiencies such as zinc deficiency are also found.

Thus, the indicators discussed above generally indicate that the overall food security situation in Egypt has improved in the post-agreement period 1995-1999 compared with the situation in the pre-agreement period. The average calorie intake increased, even though it is larger than optimally required, and protein intake has slightly improved. Food production increased at a rate that is double the rate of population increase. Self-sufficiency ratios have generally improved, particularly in cereal crops. The food import capacity ratio has been maintained at almost the same level. There is some evidence that poverty in Egypt has increased slightly, which means that although food security has improved in terms of its "availability" dimension, it has failed on its "accessibility" dimension. Thus, while agriculture has gained from the liberalization policy and technological advances, macro- and income distribution policies may have failed in maintaining the interests of the low-income population.

4.3 Food aid

Egypt used to receive a relatively large quantities of food aid, especially wheat and wheat flour provided mainly by the United States and the EU. However, food aid to Egypt has been decreasing drastically as part of a global decline in food aid. Aid in the form of wheat and wheat flour decreased from 2 million tonnes in 1990, representing about 20 percent of the total consumption in 1990, to only 20 000 tonnes in 1999, representing 0.2 percent of the total consumption in 2000. For the pre-agreement period, wheat aid averaged 723 000 tonnes but decreased to 82 000 tonnes on average for the post-agreement period.

Egypt played a leading role during the UR negotiations in advocating the inclusion of the Marrakesh Ministerial Decision as an integral part of the UR Agreements, and subsequently in calling for its effective implementation during the various meetings of the Committee on Agriculture (CoA). Egypt also provided evidence that the UR was partly responsible for the increased food import bills of the beneficiary countries, in particular because of: the decline in food aid to its lowest level in 20 years; structural changes associated with the implementation of the UR commitments; supply control measures that resulted in stock depletion; and budgetary cuts related to export subsidy reduction commitments. Egypt has also occasionally expressed disappointment with the statements of the IMF and World Bank to the CoA that the establishment of new UR-related facilities was at the present stage not justified. A number of WTO Members have endorsed the views expressed by Egypt and stressed the need for concrete action to implement the Decision.

4.4 Food security policy

Egypt is one of the largest importers of wheat in the world. It has adopted a food security policy to ensure a minimum self-sufficiency ratio for wheat in particular as this is the most important staple food in the country. The major reason behind this is the risk and uncertainty of having to purchase in the future: Will the necessary commodities be available? What will the prices be? And what is the chance that some unforeseen events (such as major purchases by another country, war or political boycott) will occur and make foodstuffs unavailable when needed? This is politically translated into a slogan raised by top officials that “Who does not have his own food, cannot be free in his own decision”.

Within this context, the GOE has taken measures that have successfully resulted in raising the self-sufficiency ratio of wheat from 25 percent in the early 1980s to 55 percent in 2000. The GOE is aiming to gain a further 10 percentage points by mixing maize flour with wheat flour to reach a self-sufficiency ratio of 65 percent.

The self-sufficiency ratios for major food products in the period 1990-1999 are provided in Table 12. The significant improvement in the self-sufficiency ratio for cereals, which increased from 61.8 percent in the pre-agreement period, 1990-1994, to 66.5 percent in the post-agreement period, 1995-2000, is clear. Wheat, which is the most important staple food, achieved the best performance among the cereals with an increase in its self-sufficiency ratio from 42.8 percent in the first period to 50.4 percent in the second period. There has also been a slight improvement in the self-sufficiency ratio of meat; from 86 to 88 percent. The self-sufficiency ratios of edible oils and sugar have undergone a slight deterioration.

Table 12. Evolution of self-sufficiency ratios of major food commodities,

Year	Wheat	Maize	Rice	Sugar	Edible oil	Meat
Av. 1990-94	42.8	74.9	110.7	63.0	13.2	86.0
Av. 1995-99	50.4	72.8	113.7	61.8	12.4	88.0

Source: Calculated from FAOSTAT.

Egypt’s population is expected to grow to 90 million by the year 2020. With economic growth at a rate of 5 percent, it is estimated that the demand for food will grow at 4 percent annually. Under these circumstances, to maintain the current self-sufficiency ratios of commodities such as wheat, edible oil, milk, meat and sugar, domestic production should increase by 4 percent annually. To achieve this goal, Egypt would need to divert more

agricultural resources to food production, leaving few of these limited resources for export production.

However, Egypt has a definite comparative advantage with specific products such as cotton and horticultural crops, especially oranges, potatoes and aromatic and medical plants. Egypt has excellent potential for the production of horticultural crops, particularly in the new land and mega-project areas, where plans are being made to develop about 30 percent of these areas for horticultural production. In effect, the competition between food products and exportable products over the available limited agricultural resource base will be determined mainly by future market access conditions. With improved market access, Egypt would better adopt a strategy that is based on allocating increasing amount of resources for high-value exportable horticultural products for which Egypt has a comparative advantage. On the contrary, restricted market access will push Egypt to adopt an import-substitution strategy that focuses on food production, which implies a less efficient use of agricultural resources. While the obvious market for horticulture is the EU, it maintains a highly protectionist trade regime in agriculture. With market access problems such as these, Egypt's traditional concerns about food security can be understood. Protectionism in the EU as Egypt's major market is not an incentive to modify its position on food security (WTO, 1999).

5 Interests and issues of concern in the next WTO round

Completion of the UR of trade negotiations has resulted in broad-based tariff reductions and easing of some important non-tariff barriers. The current round provides further opportunities for developing countries to be more active in the WTO to achieve a greater recognition of their interests. In this round of negotiations, Egypt should be concerned about the following issues: (1) the scope for increased agricultural exports; (2) impact of world price increases and potential implications for food imports; and (3) the overall effect on agricultural development in the context of a limited land and water resource base.

These issues should be added to the standard issues: market access, domestic support and export subsidies in the developed countries. The next round of negotiations will have to bring about additional market access provisions, further reductions in export subsidies, limits to quantitative restrictions, especially on agricultural trade, and more discipline in the area of trade-distorting domestic subsidies. Entry for agricultural and food goods should be no more restricted than for non-agricultural goods. In addition, Egypt has a special interest in the next round of negotiations concerning SDT, export restraints, price stability, food security, food aid and stock policies.

Based on the review and analysis in the previous sections, this section summarizes some of the issues for Egypt in further negotiations on agriculture and points to the key areas where further analysis, studies and institutional strengthening may be required as part of the preparation for these negotiations.

5.1 AOA commitments

The review in Section 2 showed that the AoA provisions on domestic support and export subsidies by and large did not circumscribe current Egyptian policies, while some difficulties were noted in the area of market access.

A thorough analysis of domestic support measures was not possible because of the lack of information on those of a trade-distorting nature (the AMS category). However, the absence of relevant information in Egypt's WTO commitments essentially implies that the AMS, if any, falls within the *de minimis* levels. This was seen in Table 5 to be the case for non-product-specific AMS, where Egypt could provide support up to roughly US$1 300 million without breaching current AoA rules, and actual box outlays were far below the *de minimis* level. For product-specific AMS, too, as regards such commodities as rice, cotton and wheat support, expenditures have probably been well within permitted limits. Recently, the GOE has eliminated the support for both rice and cotton but still maintains support for wheat for food security reasons. However, expenditure on this support was reduced after

the recent depreciation of local currency against the dollar while the same administrative farm price for procured wheat (2 million out of 6 million tonnes) was maintained.

Notwithstanding the above, rather than live with the uncertainty of its position, and possibly face inquiries by WTO members from time to time, it would be desirable for Egypt to assess its AMS levels carefully for recent years and update them periodically. The exercise would not only be useful in the WTO context but also be valuable *per se*, as part of the domestic policy analysis process.

The AoA rules on investment in irrigation is also an issue. At present, investment subsidies seem to be exempt from the reduction commitment for developing countries, although some questions have been raised in the CoA on such subsidies granted by other countries. In view of the importance of irrigation to Egypt, it is desirable for Egypt to document its own irrigation subsidies carefully and, more importantly, to follow closely the debate on this subject in WTO to ensure that the right to grant such subsidies is preserved.

On market access, Egypt's only commitment was on bound tariffs. It was seen above that its bound rates are generally lower than those of many developing countries, including most countries of the region. It is not clear whether tariffs were bound on the basis of some analysis, e.g. an examination of trends and instabilities in global commodity markets and their possible impact on importcompeting sectors. Since the next round of negotiations may result in a further reduction of bound rates, Egypt needs to analyse carefully its tariff situation, drawing upon its experience with applied tariffs over the past five years and taking into account prospective trends and instabilities in global commodity markets. It also needs to formulate a position on access to agricultural SSGs, which are particularly valuable when bound tariffs are low. Its relatively low level of tariff bindings also makes it difficult for Egypt to include many of its agricultural products in a regional trading agreement with countries that have higher tariff bindings.

5.2 Egypt's proposals in the current negotiating round

Egypt agrees with the overall objective of the new negotiating round in agriculture. However, it emphasizes the importance of the full implementation of commitments included in the current AoA. In particular, Egypt has a real interest in other countries liberalizing trade in agricultural products and in having increased market access for agricultural exports to those countries.

Market access

- Tariff levels: The tariffication process has been used by developed countries to impose high tariffs on agricultural products such as cotton and processed food, which are of great importance to Egypt in terms of exports. It is suggested that the developed countries reduce those tariffs to levels that allow real access to their markets. Developing countries should not be required to make tariff reductions, since most of those countries, including Egypt, impose extremely low tariffs on major imported food commodities.

- Tariff escalation: The developed countries impose higher tariffs on processed agricultural products such as vegetables and fruits, which Egypt produces and exports. Tariff escalation should be eliminated.

- Tariff dispersion: Some developed countries such as the EU made very little reduction on tariffs on the so-called sensitive agricultural products such as oranges and rice. It is suggested that any variation between tariffs should be reduced.

- Tariff quotas: Countries have used different quota administration arrangements, which has created problems for exporters including a lack of transparency and predictability of the market size. It is suggested that tariff quotas should be opened on a commodity-by-commodity basis, and not on an aggregate basis.

- SSG measures: The use of SSG measures by some countries means imposing additional protection, above the ordinary tariffs, on their imports of agricultural

products. Egypt requests the elimination of the safeguard embodied in the URA, as there is no justification for this procedure which is exceptional to agriculture.

Domestic support

- Unbalanced country commitments of AMS: Although the developed countries made commitments to reduce their AMS by 20 percent, some still maintain high percentages of domestic support. Developing countries like Egypt have no entitlement to AMS measures beyond *de minimis* levels or those included in the developing country box. Egypt will be looking for a substantial reduction in permitted AMS levels to a more uniform level across countries. It is suggested that developed countries should commit to reduce domestic support calculated on a commodity-by-commodity basis, and not on an aggregate basis.

- Green Box criteria: The generalization of the Green Box criteria enabled some countries to misuse it to provide domestic support leading to trade distortion. It is suggested that Green Box criteria be specified in a way that prevents support that distorts trade while allowing it for poor farmers.

Export subsidies

Egypt requests a substantial reduction in export subsidies that distort trade and weaken commitments concerning market access. Egyptian exports of some products, such as cotton and rice, have faced significant losses owing to the current domestic support in developed countries.

SPS/TBT measures

From the experience of the past five years, summarized above, it is clear that SPS/TBT measures will have an increasing influence on Egypt's exports. More needs to be done to document and share these experiences, domestic traders being best placed to know the reality. A concerted effort by the Government is required to survey these traders, analyse their recent experience and pursue unfair cases in the appropriate WTO forum, especially for Egypt's major export products, such as fruit and vegetables, both fresh and processed.

Food security

For Egypt, it is always critical to consider the food security implications of policies, including those pursued in the context of the AoA. World food price increases resulting from reduction or elimination of distorting interventions in agricultural trade have mixed impacts on the welfare of Egypt's population and economy. Farmers will be the main beneficiaries, provided that world price increases are reflected in domestic farm prices. However, large farmers with larger marketable surpluses acquire much higher gains than small farmers, whose majority production is for their home consumption, leaving very small surpluses if any. Landless agricultural workers may also benefit from the increase in agricultural prices, which most probably lead to higher agricultural wages. At the sector level, food production would be encouraged by price increases thus growing at a faster rate, leading to an improved self-sufficiency and lower food dependency. Furthermore, there will be no need for the government to provide domestic support. Also, the traditional income gap between the agricultural and non-agricultural population may be reduced, or at least stopped from deteriorating, depending on the rate of increase of agricultural prices compared with the rate of change in the prices of industrial products. In this case, income distribution may become more equal between the two sectors.

Among those losing because of world food price increases will be the government, who will have to devote more of the public budget for a larger food bill. Consumers in both urban and non-farm rural sectors will be harmed, since they will spend more of their incomes on the same amount of food. Furthermore, low- and middle-income consumers will be harmed much more than high-income consumers, since the former category devote a higher portion of their incomes to food expenditure than the latter.

Finally, the overall impact on food security of allowing free play to non-distorted agricultural prices in the country is determined to a large extent by the effectiveness of national and international trade and macroeconomic policies. Unless the government is committed to long-term macroeconomic stability, reforms in agriculture are unlikely to be effective. Overvalued exchange rates and protection for industry, for instance, can do more to reduce farmers' incentives to increase production than lower agricultural prices. In terms of international trade, Egypt and other developing countries must participate effectively in the current round of global agricultural trade negotiations and pursue better access to industrialized countries' markets. Meanwhile, industrialized countries must reduce, and eventually end, trade-distorting agricultural subsidies.

In addition to general AoA provisions, the implementation of the Marrakesh Ministerial Decision is another matter of considerable interest for Egypt, given the time and resources it has devoted in contributing to debates on the Decision so far. It has proposed the creation of a fund for the support of NFIDCs with financing from a number of international financial organization, special UN agencies, developed country donors and major exporters. The new round of negotiations provides another opportunity for working towards an effective Decision.

In the last Trade Policy Review, Egypt expressed its concern that the decision on net food-importing countries, which agreed to establish appropriate mechanisms to ensure that the impact of the UR on trade in agriculture did not adversely affect the availability of basic food on reasonable terms to developing countries, had not been implemented. Food aid levels to the net food-importing developing countries had declined at an alarming rate from 1978 to 1997, while the expectations of the affected countries in areas such as concessional finance, export credit and technical assistance had not been met.

Being one of the net food-importing countries, Egypt is a country where increased food insecurity will have to be addressed in a more systematic way than through food aid and food subsidies. There are concerns about increases in world food prices resulting from the market response to reduced subsidies and lower distortions for cereals, sugar, oilseeds and livestock products in the developed countries.

Egypt should propose schemes to dampen the impact of unstable food prices on food security, the balance of trade and government budget. In this regard, efforts should be made to include food security-related mechanisms as non-trade distorting instruments in the next round of negotiations to avoid having marketstabilizing instruments being brought to dispute settlement. The IMF and FAO have adequate programmes and facilities to deal with exceptional situations that could jeopardize the food situation in the LDCs as well as NFIDCs. Prospects to add another multilateral structure for having food import assistance should also be considered (Chaherli and Moataz, 2000).

In summary, Egypt's negotiating proposal calls for the phasing out of all forms of export subsidization over an agreed period of time. It also calls for additional strengthened rules and disciplines to prevent the circumvention of export subsidies. On domestic support, the proposal calls for reductions across the range of boxes. Countries should reduce production-distorting domestic support, and this support should be measured on a disaggregated basis. On market access, Egypt seeks the elimination of all tariffs and of other trade-distorting measures, and agrees that tariff quota administration disciplines should be strengthened.

Egypt has put forward a constructive proposal about how to address the needs of NFIDCs. As regards S&D provisions in the AoA for developing countries, Egypt's view is that it should be strengthened and expanded. This should be done in a manner that would be the least trade-distorting, while affording developing countries more flexibility to assist their agricultural sectors to compete actively in the globalizing world economy.

5.3 Research and institutional capacity

While Egypt has reasonably trained personnel and resources to assess various options with respect to WTO negotiations, Egypt ought to explore the establishment of a formal and

permanent research unit that could evaluate the impact of specific WTO commitments but also monitor future developments associated with the agreed commitments. An "International Trade Evaluation and Monitoring Unit" with representation from different ministries and agencies should be established, with four objectives: (1) evaluate trade policy-related rights and responsibilities in the context of bilateral and multilateral arrangements with regard to legal, economic and financial implications; (2) provide updated information for food security particularly with regards to seasonal and structural food gaps and policy recommendations compatible with WTO rules; (3) coordinate efforts of domestic and international bodies involved in trade in agricultural and food commodities; and (4) study the impact of trade liberalization on the domestic level and the impact on volatility in world agricultural markets.

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[31] Study prepared for FAO by Dr Gamal M. Siam. The views expressed in the paper are those of the author and do not necessarily reflect the views of the Government of Egypt.